

CHAROTAR UNIVERSITY OF SCIENCE & TECHNOLOGY

Fourth Semester of B. Tech. (EC) Examination

Nov 2012

EC 210 Analog Communication

Date: 08.11.2012, Thursday

Time: 01:30 p.m. To 04:30 p.m.

Maximum Marks: 70

Instructions:

1. The question paper comprises of two sections.
2. Section I and II must be attempted in separate answer sheets.
3. Make suitable assumptions and draw neat figures wherever required.
4. Use of scientific calculator is allowed.

SECTION – I

- Q - 1 (a) Define noise factor and derive Friss formula. [04]
- (b) Define: 1) Avalanche noise 2) Shot noise 3) Low frequency noise. [03]
- Q - 2 (a) Explain envelope detector circuit. [05]
- (b) Explain basic circuit for a BJT collector modulator. [05]
- (c) A modulating signal consists of a symmetrical triangular wave having zero dc component and peak-to-peak voltage of 11V. It is used to amplitude modulate a carrier of peak voltage 10V. Calculate the modulation index and the ratio of the side lengths L_1/L_2 of the corresponding trapezoidal pattern. [04]

OR

- Q - 2 (a) Dervie the equation for modulation index for sinusoidal Amplitude Modulation. Draw sinusoidal amplitude modulated waveforms for (a) $m = 0.5$, (b) $m = 1$, (c) $m > 1$. [06]
- (b) Calculate the thermal noise power available from any resistor at room-temperature (209 K) for a bandwidth of 1MHz. Calculate also the corresponding noise voltage, given that $R=50\Omega$. [04]
- (c) Explain Equivalent noise bandwidth. [04]
- Q - 3 (a) Explain third method of generating an SSB-SC signal. [05]
- (b) Derive the expression to generate SSB using FET balanced modulator circuit. [04]
- (c) Explain double balanced diode ring modulator. [05]

OR

- Q - 3 (a) Explain and derive equation for Effective voltage and current for sinusoidal Amplitude Modulation. [05]
- (b) Justify "Thermal noise are referred as white noise". [04]
- (c) Write a short note on pilot carrier SSB. [05]

SECTION – II

- Q - 4 (a) State and prove the scaling and time-shifting property Fourier transform. [04]
(b) Explain parse Val's theorem. [03]
- Q - 5 (a) A receiver tunes signals from 550 to 1600 KHz with an IF of 455 KHz. Find the frequency tuning range and capacitor tuning ranges for the oscillator section and for the RF section. [06]
(b) Write short note on Foster-Seeley Discriminator . [04]
(c) Derive equation for average power in sinusoidal FM. [04]

OR

- Q - 5 (a) Explain superhetrodyne principle and superhetrodyne receivers. [05]
(b) Write notes on Varactor diode frequency modulator. [05]
(c) A 15-W unmodulated carries is frequency modulated with a sinusoidal signal such that the peak frequency deviation is 6KHz. The frequency of the modulating signal is 1 KHz. Calculate the average power output by summing the powers for all the side-frequency components. [04]
Take values of The Bessel functions for $\beta = 6$ are $J_0 = 0.15$, $J_1 = 0.28$, $J_2 = 0.24$, $J_3 = 0.11$, $J_4 = 0.36$, $J_5 = 0.36$, $J_6 = 0.25$, $J_7 = 0.13$, $J_8 = 0.06$, $J_9 = 0.02$, $J_{10} = 0.01$.
- Q - 6 (a) Write a short note on Adjacent channel selectivity. [06]
(b) Explain the block diagram of FM transmitter. [06]
(c) Canadian regulations state that for FM broadcast the maximum deviation allowed is 75 KHz and the maximum modulation frequency allowed is 15 KHz. Calculate the maximum bandwidth requirements. [02]

OR

- Q - 6 (a) Explain superheterodyne receiver tracking methods. [06]
(b) Define sensitivity, gain and image rejection. [04]
(c) An AM broadcast receiver has an IF of 465 KHz and is tuned to 1000 KHz, and the RF stage has one tuned circuit with a Q of 50. [04]
(a) Find the image frequency
(b) Find the image rejection in decibels.
